Temasek POLYTECHNIC

SCHOOL OF INFORMATICS & IT

Diploma in Information Technology

Part 3: Continuous Integration/ Continuous Deployment (CI/CD)

Student Name (Matric Number)	:	Angel Zaf (2402054G)
Tutorial Group	:	P04
Tutor	:	XX
Submission Date	:	DD/MM/202X

Declaration of Originality

I am the originator of this work and I have appropriately acknowledged all other original sources used as my references for this work.

I understand that Plagiarism is the act of taking and using the whole or any part of another person's work, including work generated by AI, and presenting it as my own.

I understand that Plagiarism is an academic offence
and if I am found to have committed or abetted the offence of plagiarism in relation to this submitted work, disciplinary action will be enforced.

Note to Tutor: This page is only to be used if your subject specifically allows AI-generated content to be used. Please delete this page if it is not required.

Declaration on the use of Generative AI tools for assignments

Describe how you have used Generative AI tools such as ChatGPT or Dall.E- in your assignment.

Share the link to the conversations you had with the AI tool (i.e., the prompts you used and the responses you get from the AI tool).

Please refer to this PDF on "How to share the conversations made with ChatGPT?"



How to share the conversations made v

How do you indicate the reference?

The content generated by AI tools are not retrievable except by the user who generated them, so they are considered non-recoverable sources. Although non-recoverable data or quotations in APA Style papers are usually cited as personal communications, with ChatGPT-generated text there is no person communicating. Quoting text from ChatGPT chat is therefore more like sharing the output of an algorithm, with a reference list entry and the corresponding in-text citation.

According to the official APA Style site, ChatGPT references should be cited as:
E.g. OpenAI. (2024). *ChatGPT* (Feb 13 version) [Large language model].
https://chat.openai.com/chat

Note: The information in parentheses refers to the update or revision date of the model used. Refer to the release notes in the ChatGPT application.

Important Note:

- Do not copy answers produced by the AI tool in totality as it is considered as plagiarism.
- Do not rely on any information produced by the AI tool blindly. You should always verify the answer with other sources. Do not assume that these answers provided by the AI tool are correct.
- To achieve quality outputs from the AI tool, you should provide good prompt that is clear and specific. Be precise and provide context. Avoid asking open-ended questions.

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You are NOT allowed to modify the page margins, font sizes and line and paragraph spacings.

Please remove all prompting questions and guidelines in each section before submitting.

Report, excluding the PPMR section, should NOT exceed 8 pages.

CI/CD Pipeline Setup ----------

Build Phase

Tools/Frameworks involved:

- Jenkins
- Node.js
- Npm

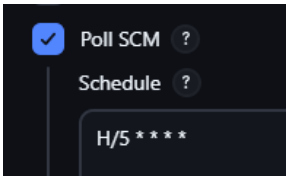
Configurations and/or Evidence:

PAT – used for Jenkins to authenticate and connect to the GitHub Repo

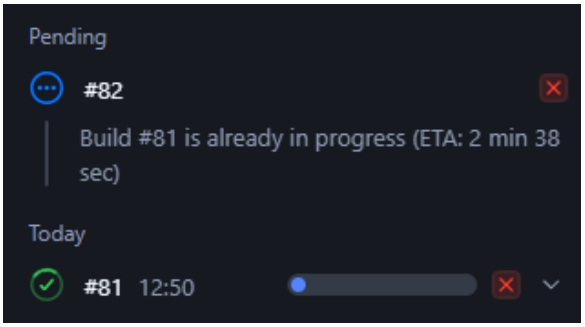
Now that Jenkins has access to GitHub, it can now pull the latest source code from the repo whenever a build is triggered automatically, allowing Jenkins to build, test, and deploy the latest version of the code without needing manual code uploads



Poll SCM – configured with H/5 * * * * so Jenkins can check the repo for changes every five minutes

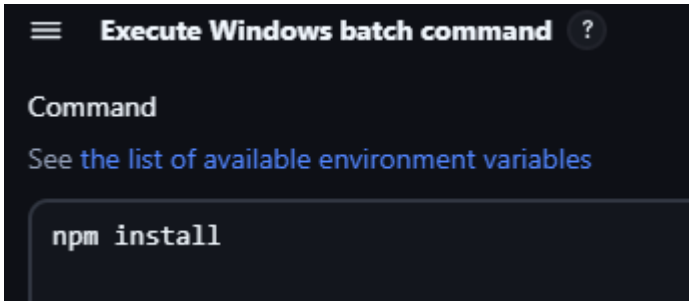


The hash based scheduling is used to stagger job execution times, preventing multiple jobs from running simultaneously and thus reduce server load



'Execute Windows batch command' is used as Jenkins is running on a Windows operating system
The build steps in Jenkins define the sequence of actions that are to be executed automatically during the CICD Pipeline

The first build step is to install project dependencies using npm install



Jenkins console output will confirm a successful build process with all dependencies installed without errors

```
C:\Users\iaman\AppData\Local\Jenkins\.jenkins\workspace\Book GURU>npm install

up to date, audited 452 packages in 3s

61 packages are looking for funding
  run `npm fund` for details

12 vulnerabilities (5 low, 5 moderate, 2 high)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force
```

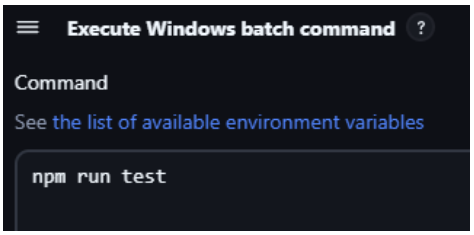
Test Phase

Tools/Frameworks involved:

- Jenkins
- Jest (backend testing)
- Supertest (API testing)
- Playwright (frontend)

Configurations and/or Evidence:

Next build step is to run automated backend tests using Jest and Supertest



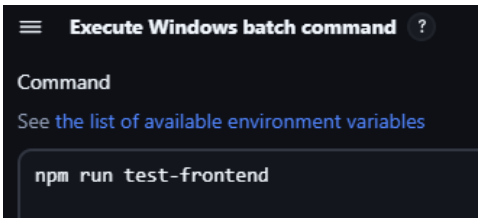
Jenkins console output will show successful executions of backend tests with all the Jest and Supertest tests cases passing

```
PASS tests/api.test.js
  Library Management API (Delete Only)
    √ DELETE /delete-book/:bookId should remove a book (49 ms)
    √ DELETE /delete-book/:bookId should return 404 if book not found (3 ms)
```

```
PASS tests/util.test.js
  Unit Tests for Delete Book Utility
    √ deletes a book successfully (200) (2 ms)
    √ returns 404 if book does not exist (bookIndex === -1) (1 ms)
    √ returns 400 when bookId is missing (1 ms)
    √ supports req.params.id fallback (1 ms)
    √ returns 404 when library.json does not exist (ENOENT) (1 ms)
    √ returns 500 on unexpected readFile error (5 ms)
    √ returns 500 if writeFile fails (6 ms)
    √ handles when parsed.books is not an array (1 ms)

-------------------------|---------|----------|---------|---------|-------------------
File                     | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-------------------------|---------|----------|---------|---------|-------------------
All files                |    80.7 |    73.68 |      50 |   80.35 |
 Book GURU               |   65.62 |     37.5 |       0 |   65.62 |
  index.js               |   65.62 |     37.5 |       0 |   65.62 | 20,37-48
 Book GURU/utils         |     100 |      100 |     100 |     100 |
  AngelDeleteBookUtil.js |     100 |      100 |     100 |     100 |
-------------------------|---------|----------|---------|---------|-------------------
Test Suites: 2 passed, 2 total
Tests:       10 passed, 10 total
Snapshots:   0 total
Time:        1.576 s
Ran all test suites.
```

Next build step is to run an automated frontend end-to-end tests using Playwright



Jenkins console output shows Playwright end-to-end frontend tests executing successfully across multiple browsers

```
Running 12 tests using 1 worker
............
  12 passed (31.1s)
[Book GURU] $ cmd /c call C:\Users\iama
```

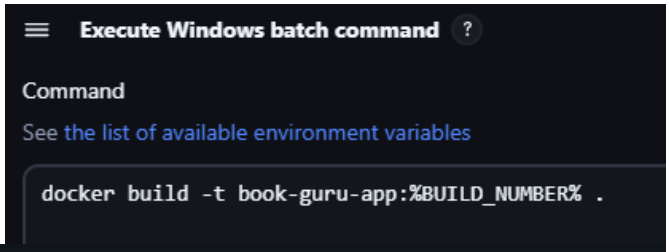
Deployment Phase

Tools/Frameworks involved:

- Jenkins
- Docker
- Minikube
- Kubernetes

Configurations and/or Evidence:

Next automated build step is to build a Docker image using a Dockerfile
The image is also tagged with the Jenkins build number so that its easy to tell which image came from which test





Name	Tag	Image ID	Created	Size	Actions
book-guru-app	81	7b3d5fa549ab	4 days ago	1.93 GB	
book-guru-app	80	972bbec7b570	4 days ago	1.93 GB	
book-guru-app	79	0efcab6ca642	9 days ago	1.93 GB	
book-guru-app	68	7beae84d18c4	9 days ago	1.93 GB	

Jenkins Console output will then show successful Docker image build for the application using the Dockerfile

```
C:\Users\iaman\AppData\Local\Jenkins\.jenkins\workspace\Book GURU>docker build -t book-guru-
app:81 .
#0 building with "desktop-linux" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 466B 0.0s done
#1 DONE 0.0s

#2 [internal] load metadata for docker.io/library/node:20
#2 ...

#3 [auth] library/node:pull token for registry-1.docker.io
#3 DONE 0.0s

#2 [internal] load metadata for docker.io/library/node:20
#2 DONE 2.3s

#4 [internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.0s
```

```
#5 [1/5] FROM
docker.io/library/node:20@sha256:3680462e16df266d30b103a3e38e2bc315bf917d0ab873fbddbee437bf0bd
2d0
#5 resolve
docker.io/library/node:20@sha256:3680462e16df266d30b103a3e38e2bc315bf917d0ab873fbddbee437bf0bd
2d0 0.0s done
#5 DONE 0.0s

#6 [internal] load build context
#6 transferring context: 1.91MB 1.3s done
#6 DONE 1.3s

#7 [2/5] WORKDIR /usr/src/app
#7 CACHED

#8 [3/5] COPY package*.json ./
#8 CACHED

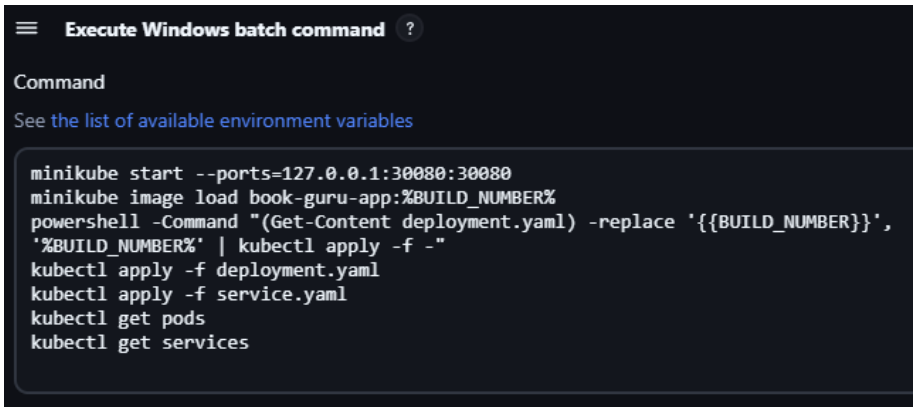
#9 [4/5] RUN npm install
#9 CACHED

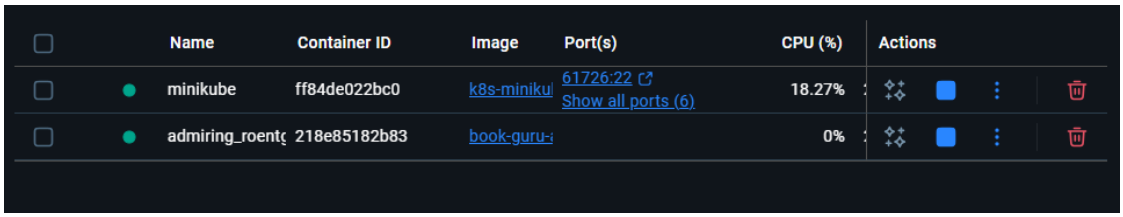
#10 [5/5] COPY . .
#10 DONE 5.1s

#11 exporting to image
#11 exporting layers
#11 exporting layers 2.6s done
#11 exporting manifest sha256:683ff21d16f3be3098bb9c1eacf13888f361deee806ff74e97f43ef4d61e069f
0.0s done
#11 exporting config sha256:c3acb971935ecaccafe08de0c18d040db4852e70447ebe3e953f4bee00018888
0.0s done
#11 exporting attestation manifest
sha256:a97bbc4dd12614dc51ce3e81252046e398255243b4b30b2d1cf20bf1fc8eeda5 0.0s done
#11 exporting manifest list
sha256:7b3d5fa549ab6b62f17934be209c4443017b2b92e01ef4db4b838da30474e18b 0.0s done
#11 naming to docker.io/library/book-guru-app:81 done
#11 unpacking to docker.io/library/book-guru-app:81
#11 unpacking to docker.io/library/book-guru-app:81 1.6s done
#11 DONE 4.3s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-
linux/vu58qc3zxucxgw8sb2p772nvq
```

Jenkins then starts a local Kubernetes cluster using Minikube and loads the Docker image into the cluster environment Kubernetes configuration files are applied using kubectl to define the application deployment and expose the service





Jenkins console logs should confirm that the Docker image was built successfully , the Minikube cluster was initialised , and the Kubernetes resources were configured

```
C:\Users\iaman\AppData\Local\Jenkins\.jenkins\workspace\Book GURU>kubectl get pods
NAME                                READY   STATUS             RESTARTS   AGE
book-guru-deploy-55f5c66c49-rzcjm   0/1     InvalidImageName   0          3h11m
book-guru-deploy-68cc86cf49-j4tlk   0/1     ContainerCreating  0          1s
book-guru-deploy-68cc86cf49-q7ms2   0/1     Terminating        0          3h11m
book-guru-deploy-6bfd689f85-dtqmt   0/1     Terminating        0          4d22h
book-guru-deploy-7bfb76655d-krl9m   0/1     ContainerCreating  0          1s

C:\Users\iaman\AppData\Local\Jenkins\.jenkins\workspace\Book GURU>kubectl get services
NAME               TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
book-guru-service  NodePort    10.103.229.226   <none>        5050:30080/TCP   7d
kubernetes         ClusterIP   10.96.0.1        <none>        443/TCP          7d

C:\Users\iaman\AppData\Local\Jenkins\.jenkins\workspace\Book GURU>exit 0
No emails were triggered.
Finished: SUCCESS
```

Challenges Faced

I did follow the lab thoroughly , so I didn't come across much challenges in terms of getting my build to run successfully
Most of the errors were a result of either not running my Docker containers or a misspelling

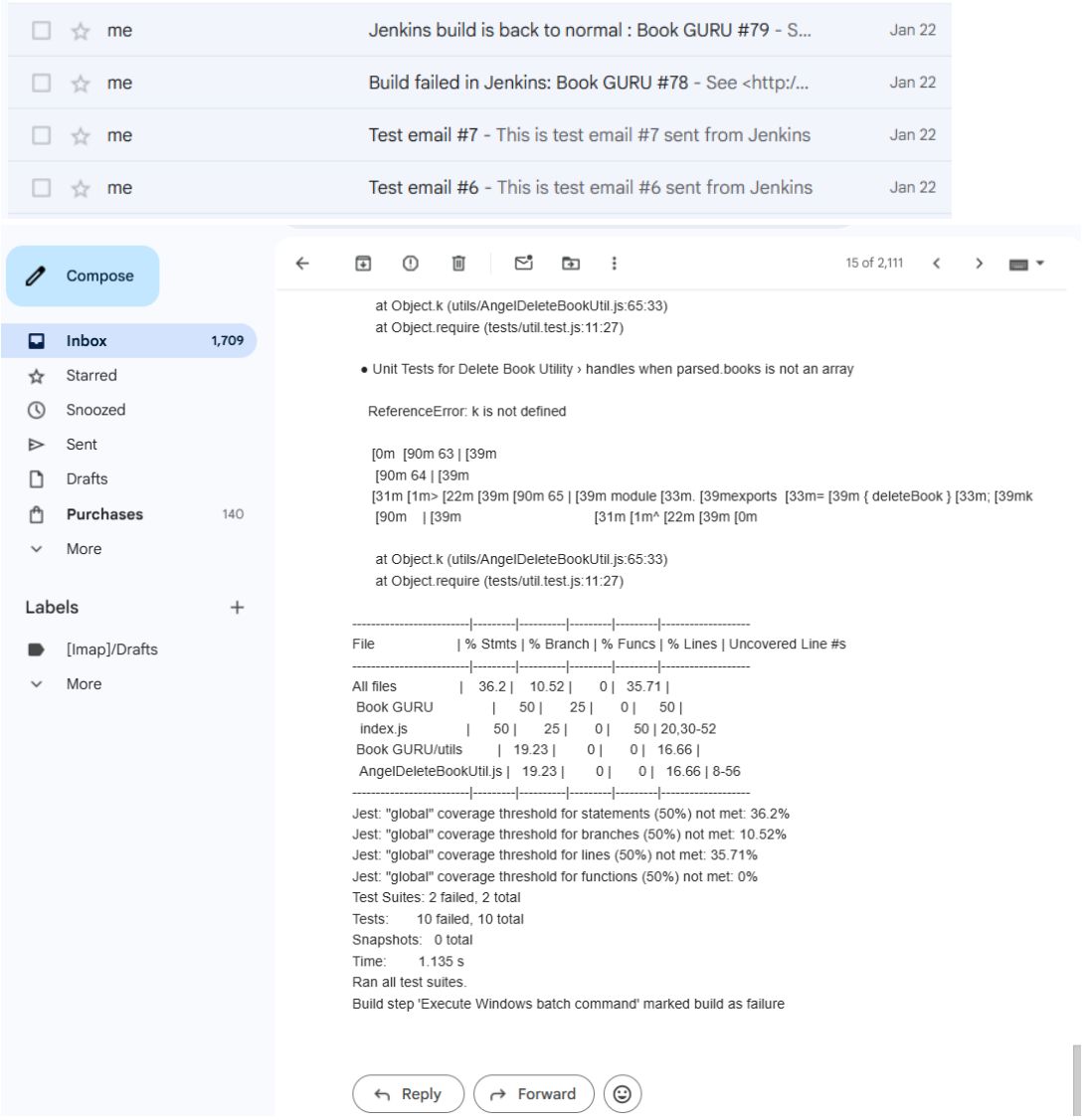
Additional Features ----------

Jenkins Email

Purpose

Jenkins Email Notification was implemented to enhance the monitoring and feedback within the CI/CD pipeline. This feature automatically notifies selected emails of pipeline events such as the build failures and the successful recovery after a failed build

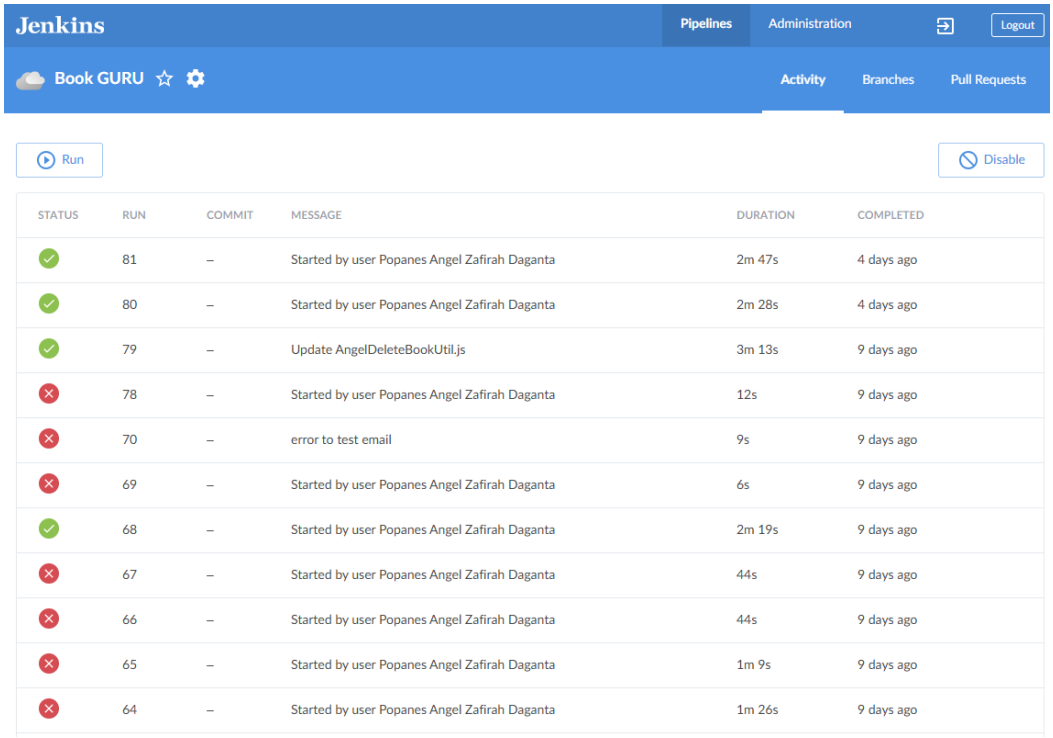
By providing immediate feedback, it reduces the time required to identify and respond to issues in the pipeline



Blue Ocean

Purpose

Blue Ocean was implemented to improve visualization and monitoring of the CI/CD pipeline It provides a more clearer interface for viewing the pipeline stages , build status , and execution progress , making it easier to understand the flow of the pipeline
This provides better monitoring , faster debugging , and improved pipeline management



STATUS	RUN	COMMIT	MESSAGE	DURATION	COMPLETED
✓	81	–	Started by user Popanes Angel Zafirah Daganta	2m 47s	4 days ago
✓	80	–	Started by user Popanes Angel Zafirah Daganta	2m 28s	4 days ago
✓	79	–	Update AngelDeleteBookUtil.js	3m 13s	9 days ago
✗	78	–	Started by user Popanes Angel Zafirah Daganta	12s	9 days ago
✗	70	–	error to test email	9s	9 days ago
✗	69	–	Started by user Popanes Angel Zafirah Daganta	6s	9 days ago
✓	68	–	Started by user Popanes Angel Zafirah Daganta	2m 19s	9 days ago
✗	67	–	Started by user Popanes Angel Zafirah Daganta	44s	9 days ago
✗	66	–	Started by user Popanes Angel Zafirah Daganta	44s	9 days ago
✗	65	–	Started by user Popanes Angel Zafirah Daganta	1m 9s	9 days ago
✗	64	–	Started by user Popanes Angel Zafirah Daganta	1m 26s	9 days ago

Plan-Perform-Monitor-Reflect (PPMR) Journal ----------

PLAN

<table>
<tr><td colspan="2">Monitoring Tool:
Jenkins Email Notification</td></tr>
<tr><td colspan="2">Ideas/Issues (Topic knowledge)</td></tr>
<tr><td>What I know
Any prior knowledge that will be useful for this task?

After learning about the Jenkins pipeline , I understand that monitoring builds and the test results are important in a CI/CD workflow</td><td>What I don’t know, or
What I need to find out/work on
What do you think are some of the knowledge that you will need to acquire to perform the task?

I need to learn how to configure the Jenkins Email Notifications
- How to set up SMTP authentication
- Select which triggers the notifications
- Ensure emails are sent automatically after each build</td></tr>
<tr><td colspan="2">Action (Identify resources/information)</td></tr>
<tr><td>Get help from
Who are you intending to seek help from?

I intend to clarify any doubts with my lecturer</td><td>Resources I should use
What do you think are some of the possible resources that you can make use of (e.g. YouTube, official website, guided labsheet in LMS, etc)?

Youtube</td></tr>
<tr><td colspan="2">Time Management</td></tr>
<tr><td colspan="2">By when must I complete
Breakdown your task into bite sizes to help yourself manage this SDL task better. (e.g. Research on XX, Research on YY, etc). Set yourself some targets! Indicate deadlines for each sub tasks that you have identified.

20 jan , finish the labs and the CI/CD pipeline , so I can start implementing additional features
22 jan , implement Blue Ocean by following lms
23 jan , implement another additional feature (Jenkins email notif)
Finish by 26 jan , so I can consult my lecturer during the makeup lessons on 27 jan should I have any problems</td></tr>
</table>

PERFORM

Strategies that I am using to
1. Monitor progress/keep on task (is the task taking longer than expected?) *Share some strategies that you have used to ensure that you are on track* The task took a lot quicker than expected , allowing me more questions to ask my lecturer during classes for clarification on what the error means and advice on how to fix them
2. Locate resource/get help (where & how) *Share some strategies that you have used to efficiently locate resources/ get help* I first googled how to install the Jenkins Email Notification , and clicked the first youtube video that popped up
3. Check understanding (understand everything I need to understand) *Share some strategies that you have used to confirm your understanding* I used the official Jenkins website to make sure I understood what I was doing https://plugins.jenkins.io/email-ext/
4. Motivate myself *Share some strategies that you have used to motivate yourself to perform the SDL task.* I told myself to get it done before 27 jan so I could have my lecturer analyse any errors I may potentially run into, and have him not only teach me how to tell the error, but advice me on how to fix it as well
Resources/Information I have gathered
List down the online resources or books reference you have identified for the fulfillment of the SDL task. this youtube video https://www.youtube.com/watch?v=j7jmXYZK5dU

MONITOR

Evaluating Time Management & Potential Information Sources
Tasks Monitoring • *Describe how you monitor your own progress while researching on the chosen domain* I tracked my progress by using the builds , whether they were successful or not , and whether I was making progress
Resources Monitoring • *Are the resources reliable and level appropriate for the assignment?* The youtube tutorials were really helpful and simple to understand , which I find appropriate for this assignment As for reliability , I did use the Jenkins official web to check about email notification , so it was reliable there And when I encountered a problem , my lecturer was also really reliable person to consult for help

REFLECT

Write a reflection on your learning
Articulated Learning • *Write about some strategies or skills you have acquired while working through the assignment.* • *Do you think you have progressed well? Did you have better time management?* • *Given another assignment/project, might there be anything you would do differently from what you have done?* • *How would you rate yourself as a Self-Directed Learner?* Through this assignment , I acquired stronger problem-solving skills , specifically like configuring Jenkins features such as email notification , automated testing , and deployment steps I also improved my ability to read and interpret the console output logs , which I feel has helped me identify the root causes of why the build failed I think I progressed well and had better time management , and also learnt to make sure I had extra time before the deadline in case I encounter any errors and may need to consult my lecturer If given another assignment , I think I should make it a point to document any errors I come across along the way instead of trying to solve it immediately , so I can bring it up during another consultation session to make sure I never make that mistake again If I were to rate myself , I think I would say im quite capable of being a self-directed learner . Since I have access to youtube and other materials , I find myself confident in being able to learn by myself
Evaluation The Plan-Perform-Monitor-Reflect (PPMR) process has helped me to challenge myself towards achieving the desired standards I set for my assignment. ☐ Strongly agree ☐ Agree ☐ Disagree ☐ Strongly disagree